



PublisherStudio documentation

Version 2.2.3

Curated product, architecture, engineering, and reference handbook with a complete linked inventory of every generated API page.

HTML-rendered, searchable, and packaged with the same documentation site used in the application and on GitHub Pages.

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PublisherStudio documentation

Version 2.2.3

PublisherStudio is a local-first publishing workspace for page layout, stories, spreadsheets, pictures, audio, video, streaming, interactive content, and self-contained exports.

This site uses the same documentation shell as LocalGPT, so navigation, themes, side rails, and reference pages behave consistently. 🐾

Choose a path

Use PublisherStudio

Create a publication and learn the workspace, page, object, story, and ribbon model.

[Open the user guide](#)

Build rich content

Work with stories, spreadsheets, pictures, audio, video, animation, streaming, and recording.

[Explore pictures and media](#)

Build and maintain it

Read the architecture, development requirements, installer behavior, and documentation pipeline.

[Open engineering guidance](#)

Look up details

Browse export behavior, privacy boundaries, the optional LocalGPT connection, and the API reference.

[Browse reference material](#)

Architecture at a glance

Human authorBlazor + DevExpress workspaceApplication servicesPublication modelMedia and streamingExport pipelines

The editable publication remains authoritative. Browser runtimes accelerate interaction and rendering, while C# services own project state, validation, persistence, export, and approved external connections.

Complete documentation set

The same themed HTML site is shipped inside PublisherStudio and published to GitHub Pages.

The packaged PDF follows the same readable portrait handbook model as LocalGPT: it contains every maintained PublisherStudio chapter plus a complete linked namespace/type inventory. Detailed API members stay in the searchable HTML site, GitHub Pages, and the in-app viewer.

 [Download the Kawaii handbook](#)

Animation and interaction

Animations belong to the publication model, not to one export format.

Each object can own entrance, emphasis, motion, media, and exit steps. Pages can own transitions, and objects can respond to clicks or presentation events.

Timeline rules

Animation order is page-wide. This gives the timeline one deterministic sequence across text, pictures, shapes, media, panels, and data visuals.

Presentation actions

Interactions can navigate pages, open safe URLs, show or hide targets, replay animations, and control media. An object can be hidden at presentation start while remaining visible and editable in the authoring workspace.

Export mapping

The HTML runtime maps semantic animation records to browser animations. Print and static images show the authored visual state. Video export renders the timeline. Unsupported behavior should be reported as a capability or loss marker rather than silently disappearing.

Architecture

PublisherStudio is a dependency-injection-oriented modular monolith. The design keeps the application easy to ship while giving each responsibility one clear home. 🌸

Main roots

- **Components** own UI state and user interaction.
- **Controllers** own backend HTTP request entry points and results.
- **Hubs** own persistent connection entry points.
- **HostedServices** own application-lifetime scheduling.
- **Services** own reusable processing, persistence, devices, files, networks, FFmpeg, and operating-system work.
- **BusinessObjects** own authoritative documents, requests, state, and result contracts.

Reusable work flows toward Services and BusinessObjects. Services do not depend on Components, Controllers, Hubs, or HostedServices. Specialized use cases stay in a UseCases subnamespace below their owning service area rather than becoming a competing application root.

```
flowchart LR
    UI[Blazor Components] --> S[Reusable Services]
    C[Controllers: backend request entry] --> S
    HB[Hubs: persistent connection entry] --> S
    HS[HostedServices: lifetime work] --> S
    S --> BO[BusinessObjects]
```

Service-owned behavior

Program.cs is the host composition root. Runtime state and reusable behavior belong to DI-owned services. PublisherStudio does not add application convenience statics to bypass composition.

P/Invoke and native exports belong behind injected lifetime services. Records, DTOs, constructors, and other BusinessObjects describe data; they do not acquire files, devices, networks, loggers, or host state.

Canonical content

The native publication model is authoritative. External formats are adapters:

```
external input → parse → temporary canonical model → validate → commit
canonical model → capability analysis → map → external output
```

Every semantic contract has one authoritative declaration. Service-local copies of BusinessObjects are not allowed.

Browser boundary

C# owns canonical state. JavaScript owns high-frequency browser interaction and commits bounded results. Listeners, observers, object URLs, pointer capture, and media streams are released on rebind or disposal.

Editor interaction overlays live in a local positioned stacking context. They never change publication layer order, and a local editor problem is not solved with an application-wide z-index.

Safe input boundaries

XML importers prohibit DTD processing and external resolvers. Archives validate paths before extraction. New packages, native binaries, or helper processes require an explicit architectural reason and release review.

Developer build

Requirements

- .NET 10 SDK
- Windows PowerShell 5.1 or newer for the maintained scripts
- Node.js for preparing licensed DevExpress browser assets and documentation PDF generation
- A licensed DevExpress build identity for preparing runtime assets

First source build

```
.\Prepare-DevExpressAssets.cmd  
.\Build-LocalDevelopment.ps1 -Configuration Debug
```

The preparation command restores the pinned browser packages and generates the public runtime license. The private DevExpress license remains on the build machine.

Release build

```
.\Build-Release.ps1 -Runtime win-x64
```

All supported runtimes can be built sequentially with:

```
.\Build-AllRuntimes.ps1
```

The release lane restores the authoritative LocalGPT wire protocol package, publishes the application and standalone setup, validates configuration and documentation, then creates manifest-backed archives.

Quality gates

The maintained build checks architecture, diagnostics, localization, InteractiveServer render modes, async continuations, component boundaries, text ownership, iterator policy, system-variable initialization, JavaScript diagnostics, publish profiles, installer workflow, XML documentation coverage, and documentation output.

Release record

Each release keeps one concise changelog and one validation report. Old work diaries and personal progress notes are not part of the public documentation. Completed, partial, and deferred work is summarized only when it helps users or maintainers understand the shipped state.

Documentation system

PublisherStudio documentation follows the same release shape as LocalGPT while using PublisherStudio names, routes, and source layout.

Inputs

- maintained Markdown under docs/articles;
- public and protected C# XML comments from PublisherStudio.Web.xml;
- the built PublisherStudio.Web.dll for API metadata.

Outputs

- searchable DocFX HTML under wwwroot/help-docs;
- PublisherStudio-<version>.pdf;
- generated API pages;
- documentation-status.json;
- a searchable XML-comment catalog exposed by the running app.

Kawaii design guide

The website and PDF use the same soft pink, violet, and warm neutral palette. Decorative motion respects reduced-motion preferences. Light, dark, and system themes are selectable from a dedicated control that remains above the navigation layer.

The decoration is a smile, not a fog machine: headings stay clear, code remains readable, and technical warnings keep their seriousness.

GitHub Pages

The workflow publishes the documentation already shipped inside a release archive. It validates the manifest, theme assets, relative links, API index, and optional PDF before deploying. This keeps the public site tied to a real PublisherStudio release rather than an unrelated documentation-only build.

Editor workspace

The editor keeps page structure, object order, and reusable tools in one view.

Main areas

- **Title bar** — publication name and save state.
- **Ribbon** — create, arrange, animate, stream, export, and open help.
- **Pages pane** — page selection and ordering.
- **Mainframe** — the active publication page.
- **Inspector** — properties for the selected page or object.
- **Timeline** — animation and media timing when enabled.

Select and arrange objects

Click an object to select it. Use Ctrl or Shift for multi-selection. The Arrange commands move objects through the page layer stack, align them, or group them.

Snapping can use the grid, guides, page edges, and other objects. Rulers and guides are workspace aids; they are not exported as publication content.

Interaction ownership

A gesture has one owner. The Mainframe owns ordinary page gestures. Picture, media, panel, and spreadsheet studios own their local editing gestures while open. Temporary handles, masks, and playheads never become publication layers.

Safe cancellation

Closing or cancelling a studio leaves canonical page placement and layer order untouched unless you explicitly applied a result. This separation is one of the reasons the editor can support many object types without each tool inventing its own page model.

Getting started

PublisherStudio opens directly into the publication workspace. A new document contains a page, a canvas, a ribbon, a page list, and a properties panel.

Your first little publication

1. Give the publication a clear name in the title bar. 2. Choose **Insert** and add a text box, picture, spreadsheet, panel, or media object. 3. Move and resize the object on the page. 4. Use the right-hand inspector to adjust the selected object. 5. Save the native project before exporting. 6. Choose **File** and create a PDF, image, website, or recorded presentation.

The native project is the source of truth

PublisherStudio stores editable publication data in its own project format. Exported images, websites, PDFs, and videos are delivery formats. Keep the native project when you want to edit the work later.

Choose the application language

Use the language selector in the upper-right corner of PublisherStudio. The choice reloads the current local page, stores a request-culture cookie, and applies one complete JSON catalog to the application shell.

Application language and publication language are separate: the global selector changes PublisherStudio, while the Inspector language setting controls exported document and website metadata.

Local by default

The application listens on the local loopback interface. It does not need LocalGPT to create or publish documents. The optional LocalGPT page remains quiet until you choose to connect it.

Note

The first source build needs the prepared DevExpress browser assets. End-user release packages already contain the runtime assets they need.

Installer and updates

PublisherStudio uses the same simple deployment shape as LocalGPT. The setup is a one-click operation and keeps every runtime beneath one product-owned AppData folder.

Installed layout

The installation root is:

```
%LOCALAPPDATA%\PublisherStudio
```

Application and setup releases keep their runtime wrappers inside that root:

```
PublisherStudio\  
  winx64\  
    PublisherStudio.Web.exe  
  setupwinx64\  
    PublisherStudio.Setup.exe  
    Install.cmd  
    Update.cmd  
    Start.cmd
```

The wrapper names vary by operating system and architecture, but the product root does not move to Programs and does not reuse the former BlazorPublisher application folder.

One-click behavior

Double-clicking `PublisherStudio.Setup.exe` performs the normal install/update path:

1. download the matching application release; 2. extract it into `%LOCALAPPDATA%\PublisherStudio`; 3. download and refresh the matching setup release; 4. ensure FFmpeg is available; 5. create the required Desktop and Start Menu shortcuts; 6. start PublisherStudio on port 58071.

No command-line argument is required.

Updates

Normal updates extract reviewed release files over the existing product root. The root is not deleted unless `--force-delete` is explicitly supplied. Files that are not part of the incoming release remain untouched.

When setup is started from its installed setup folder, it continues from a temporary copy before extraction. This lets the installed setup executable and launchers be replaced during the same one-click update.

Required shortcuts

Both Desktop and Start Menu receive these entries:

- **PublisherStudio Install;**
- **PublisherStudio Update;**
- **PublisherStudio Start;**
- **PublisherStudio Folder.**

The Install, Update, and Start entries call the checked-in command files from the matching setup runtime folder. The Folder entry opens %LOCALAPPDATA%\PublisherStudio directly.

Compatibility

Former `--*-blazorpublisher` command-line names remain accepted so an older launcher can invoke the repaired setup. New launchers use the PublisherStudio names.

LocalGPT and 1-Wire

PublisherStudio works without LocalGPT. The 1-Wire connection is an optional local collaboration path.

Discovery and connection

PublisherStudio listens for compact LocalGPT discovery broadcasts on UDP port 51141. A selected peer can then establish the larger approved transport over TCP port 51140.

Discovery does not grant control. The user must approve the link, and each protected eye, hand, screen, spreadsheet, media, or publishing capability remains subject to its permission policy.

Default ports

- PublisherStudio web host: 58071
- LocalGPT 1-Wire TCP: 51140
- LocalGPT discovery UDP: 51141

These contracts stay separate so both applications can run side by side.

Protocol ownership

The shared wire protocol is versioned independently and consumed from the authoritative `LocalGPT.WireProtocolVersion` package. PublisherStudio does not carry a second copy of the protocol project.

Quiet standalone behavior

When LocalGPT is absent, PublisherStudio continues normally. The Organic Plugins page reports that no peer is connected instead of treating the optional capability as an application failure.

Runtime identity and small devices

PublisherStudio identity and secrets are generated at runtime; they are not shipped as a shared default credential. Small peers such as an ESP32 can participate only through an approved capability and transport profile. Discovery alone never grants an ESP32—or any other peer—permission to capture, publish, or control content.

Pictures and media

Picture Studio and Media Studio keep editing details inside the selected publication object.

Picture Studio

Picture documents can contain raster, text, shape, fill, and procedural layers. Crop, masks, transparency, filters, tint, blend modes, drawing, and polygon selections remain editable until you apply the result.

PublisherStudio stores both a rendered image and the editable picture document. The rendered image keeps normal publication export simple; the picture document lets you return for non-destructive edits.

Video and audio

Media objects use ordered segments. You can trim, split, merge, copy, replace, and arrange segments without changing the object's page position or layer order.

Video regions are stored as normalized source coordinates, so they remain aligned across different source sizes. Audio uses time only and does not pretend to have a two-dimensional region.

Recording preview

The live preview keeps the active browser media stream attached to the preview element. If Blazor replaces the video element, a bounded watchdog reattaches the stream. Saved recording data stays separate from the preview surface.

Local editing overlays

Internal media sections and selection overlays never become page siblings or picture layers. They are transient editor projections inside a local positioned stacking context. The publication object remains the canonical content, while browser-side pointer feedback stays lightweight and disposable.

Open picture and publication interchange

PublisherStudio keeps editable source information when it can and reports losses when a target format cannot represent a feature. Supported adapter families include SVG / SVGZ, OpenRaster, OpenDocument Drawing, and OpenDocument Presentation. The native PublisherStudio project remains the complete editable source.

Open video projects

Project format is not a media codec or container. PublisherStudio can inspect and import project structures from OpenTimelineIO, MLT XML, Kdenlive, Shotcut, XGES, OpenShot, and CMX 3600 EDL. OBS Scene Collection data may be used as source context where supported.

The first editable projection is the active video-track projection, not full multitrack compositing. Missing media is reported for relinking instead of silently discarded, and the original project remains the safest archival copy. 📺

Audio interchange can preserve production metadata through Broadcast WAV where the selected import or export path supports it.

Privacy and security

PublisherStudio is local-first, but local software still needs careful boundaries.

Data ownership

Publication projects, recordings, workbooks, pictures, settings, and protected connection state remain under the signed-in user's account. Exporters do not upload content by default.

Untrusted input

Imported archives, SVG/XML, websites, media, JSON, spreadsheet files, LocalGPT envelopes, and browser messages are treated as untrusted. Importers validate paths, sizes, required files, and dangerous content before committing changes.

Credentials

OAuth sessions, stream keys, LAN secrets, and machine-specific capture settings are stored outside publications and templates. They are not copied into normal interchange exports.

Loopback host

The default web endpoint binds to `127.0.0.1`. Opening PublisherStudio to another interface is a deliberate deployment decision and should be paired with appropriate authentication, firewall, and transport controls.

Publishing and export

PublisherStudio can deliver one project in several forms.

Print and PDF

Print uses the publication page model and browser print surface. Choose **Print / Save as PDF** when the target is a fixed document.

Images and SVG

PNG and JPEG are raster outputs. SVG keeps vector-friendly content where possible and freezes or marks media that cannot remain interactive.

Websites

- **Interactive presentation HTML** keeps page navigation and interaction.
- **Single-file website** embeds the delivery payload into one HTML file.
- **Structured website ZIP** creates a normal file tree with content-addressed assets and the PublisherStudio runtime.

The structured exporter validates archive paths, preserves original media when optimization fails, and does not upload project data.

Recorded presentation

Video export renders publication timing into a media file. Keep the native publication beside the export so the sequence can be revised later.

Tip

Publish for the audience, but save for your future self. The native project is the editable keepsake. 🎀

Structured website media policy

Structured websites avoid Base64 overhead by writing normal content-addressed assets. Preserve source media whenever possible: the exporter prefers to keep it when the browser can use it safely. Optional conversions may target PNG, WebP, AVIF, WebM, or lossless FFV1 workflows, depending on the source and available tooling.

Browser conversion is coordinated through Blazor and JavaScript interop. FFmpeg.wasm can be an optional browser-side helper, but export does not depend on it: when optimization is unavailable or fails, PublisherStudio keeps the compatible source instead of damaging the publication.

Stories and spreadsheets

PublisherStudio embeds rich documents and workbooks as normal publication objects.

Rich stories

Story editing uses DevExpress RichEdit. A story can keep formatted text, page settings, fields, backgrounds, and document content inside the publication.

Apply saves the edited story back to the selected text frame. Cancel leaves the existing object unchanged.

Spreadsheets

Spreadsheet Studio uses the DevExpress ASP.NET Core Spreadsheet in a same-origin editor surface. Workbooks can be created or imported, edited, downloaded, and applied back to a spreadsheet frame.

The page canvas and exports use a safe preview of the workbook. Executable workbook content is not injected into the publication DOM.

Practical tips

- Keep source workbooks when macros or unsupported features matter.
- Use **Fit** when a whole sheet should remain visible.
- Use **Clip** when the frame is a deliberate viewport.
- Reopen a spreadsheet frame to continue editing the embedded workbook.

Streaming and recording

Streaming Studio combines scenes, live inputs, chat, hotkeys, recording, and output control in one local session.

Session flow

1. Open Streaming Studio. 2. Choose or create a profile. 3. Add page, camera, screen, window, audio, or browser sources. 4. Run a dry test. 5. Start recording or a configured output. 6. Stop the output before closing the session.

Local capture

Browser capture is used where it fits. Windows process-loopback capture and FFmpeg services are selected only on supported platforms. Credentials and stream keys stay in protected local stores, not inside publications or interchange files.

Preview and saved media

The preview is a live projection. The recording pipeline owns the saved bytes. A preview that needs reattachment should not corrupt a complete recording.

When the preview turns black

First check whether the saved recording is still complete. If it is, reopen the preview or change the active source. The diagnostics bridge records browser-side failures, while expected circuit disconnects remain low-noise diagnostics.

Troubleshooting

The source builds but the browser UI is incomplete

Run `Prepare-DevExpressAssets.cmd`. A clean source package intentionally omits generated licensed DevExpress runtime assets.

The application reports a port override warning

PublisherStudio uses the configured Kestrel endpoint as the authority. The maintained default is `127.0.0.1:58071`. Remove conflicting `ASPNETCORE_URLS` or launch-profile addresses when you want a quiet startup log.

LocalGPT is not discovered

Confirm that both applications are running, UDP port 51141 is available, and the firewall allows local discovery. PublisherStudio remains usable while discovery is unavailable.

A browser circuit disconnects

`OperationCanceledException` and `JSDisconnectedException` can occur while a tab reloads or closes. They should be logged as expected disconnect diagnostics, not presented as data-loss errors. Reopen the page and check the application log for a preceding operational failure.

Recording preview becomes black

Check the saved recording first. If the file is complete, reopen or reselect the preview source. The preview-rebind watchdog is designed to repair a replaced video element without changing the saved recording pipeline.

Documentation is missing

A normal release build generates `wwwroot/help-docs`. The in-app Documentation page shows HTML, PDF, and XML-comment availability. Developer builds can diagnose the documentation lane separately with the build properties described in [Documentation system](#).

PublisherStudio API reference

This source-backed API preview exposes the full maintained namespace and public-type inventory. Owner builds replace every type page with the complete DocFX member reference.

PublisherStudio2 public types	PublisherStudio.BusinessObjects287 public types
PublisherStudio.BusinessObjects.Diagnostic s1 public types	PublisherStudio.BusinessObjects.Streaming 7 public types
PublisherStudio.Components.Editor5 public types	PublisherStudio.Components.Shared1 public types
PublisherStudio.Controllers20 public types	PublisherStudio.Controllers.Streaming.UseC ases6 public types
PublisherStudio.Diagnostics3 public types	PublisherStudio.HostedServices.OrganicPlu gins1 public types
PublisherStudio.HostedServices.Streaming2 public types	PublisherStudio.Hubs.Streaming.Chat1 public types
PublisherStudio.Hubs.Streaming.Lan1 public types	PublisherStudio.Models1 public types
PublisherStudio.Services47 public types	PublisherStudio.Services.Automation10 public types
PublisherStudio.Services.CodeEditing4 public types	PublisherStudio.Services.Configuration19 public types
PublisherStudio.Services.Documentation4 public types	PublisherStudio.Services.MediaConversion2 public types
PublisherStudio.Services.MediaStudio.UseC ases1 public types	PublisherStudio.Services.OpenScad16 public types
PublisherStudio.Services.OrganicPlugins29 public types	PublisherStudio.Services.Panels1 public types

PublisherStudio.Services.PictureStudio.Import2 public types

PublisherStudio.Services.Publication2 public types

PublisherStudio.Services.Publication.Import1 public types

PublisherStudio.Services.Streaming.Capture11 public types

PublisherStudio.Services.Streaming.Chat1 public types

PublisherStudio.Services.Streaming.Encoding8 public types

PublisherStudio.Services.Streaming.Hotkeys5 public types

PublisherStudio.Services.Streaming.Lan4 public types

PublisherStudio.Services.Streaming.MediaHost4 public types

PublisherStudio.Services.Streaming.Metadata1 public types

PublisherStudio.Services.Streaming.OAuth1 public types

PublisherStudio.Services.Streaming.Profiles1 public types

PublisherStudio.Services.Streaming.Session8 public types

PublisherStudio.Services.Streaming.Settings1 public types

PublisherStudio.Services.Streaming.UseCases.Chat3 public types

PublisherStudio.Services.Streaming.UseCases.Editor1 public types

PublisherStudio.Services.Streaming.UseCases.Ingest1 public types

PublisherStudio.Services.Streaming.UseCases.Lan3 public types

PublisherStudio.Services.Streaming.UseCases.NativeCapture1 public types

PublisherStudio.Services.Streaming.UseCases.Runtime2 public types

PublisherStudio.Services.Streaming.UseCases.Sessions1 public types

PublisherStudio.Services.UserExperience2 public types

PublisherStudio.Services.VideoStudio.Export6 public types

PublisherStudio.Services.VideoStudio.Import1 public types

Complete API page inventory

590 generated API pages are grouped by namespace and kind. Each name links to the matching GitHub Pages HTML reference; the packaged HTML site remains the detailed offline reference.

PublisherStudio

Namespace (1)

[PublisherStudio](#)

Type (2)

[Program](#)

[StreamingServiceCollectionExtensions](#)

PublisherStudio.BusinessObjects

Namespace (1)

[PublisherStudio.BusinessObjects](#)

Type (287)

[ApiSurfaceDescriptor](#)

[ApiSurfaceMethodDescriptor](#)

[AudioElement](#)

[AutomationCompletion](#)

[AutomationRequestStatus](#)

[BarcodeElement](#)

[BrowserAutomationCommand](#)

[BrowserAutomationCommandKind](#)

[BrowserMediaDeviceInfo](#)

[BrowserRuntimeTemplateRequest](#)

[BrowserScreenshotRequest](#)

[BusinessObjectContextSnapshot](#)

[BusinessObjectDescriptor](#)

[CanvasInsertRequest](#)

[CartesianChartStyle](#)

[CodeCommentRequest](#)

[CodeLanguageProfile](#)

[CodeTextRequest](#)

[CodeTextResult](#)

[CodeTokenSpan](#)

[ConnectorAnchor](#)

[ConnectorDashStyle](#)

[ConnectorElement](#)

[ConnectorEndpoint](#)

[ConnectorEndpointKind](#)

[ConnectorMarker](#)

[ConnectorPathKind](#)

[ConnectorToolKind](#)

[DataBubblePoint](#)

[DataChartPoint](#)

[DataFinancialPoint](#)

[DataPiePoint](#)

[DataRangePoint](#)

[DataSankeyPoint](#)

[DataSparkPoint](#)

[DataTreeMapPoint](#)

[DataVisualAggregationMode](#)

[DataVisualArgumentMode](#)

[DataVisualElement](#)

[DataVisualKind](#)

[DataVisualSortMode](#)

[DevExtremeComponentElement](#)

[DocumentObjectDataScope](#)

[ExternalFileDropRequest](#)

[FillPictureLayer](#)

[GuideLine](#)

[GuideOrientation](#)

[HtmlEmbedElement](#)

[ImageBlendMode](#)

[ImageFrameElement](#)

[ImageMaskShape](#)

[ImageTintMode](#)

[InterchangeIssue](#)

[InterchangeIssueSeverity](#)

[LiveSourceElement](#)

[MeasurementUnit](#)

[MediaConversionCapabilities](#)

[MediaConversionInsertRequest](#)

[MediaConversionJobInfo](#)

[MediaConversionJobStatus](#)

[MediaConversionOptions](#)

[MediaConversionPreset](#)

[MediaConversionProfile](#)

[MediaConversionScaleMode](#)

[MediaConversionTarget](#)

[MediaEditorResult](#)

[MediaFramePoint](#)

[MediaProjectMarker](#)

[MediaSourceInfo](#)

[MediaSourceReference](#)

[MediaStudioMouseMode](#)

[MediaTemporalSection](#)

[MediaTimelineTrack](#)

[MediaTimelineTrackKind](#)

[MediaTimelineTransition](#)

[MediaTimelineTransitionKind](#)

[NativeMediaDeviceInfo](#)

[OpenScadAnimationEasing](#)

[OpenScadAnimationProperty](#)

[OpenScadAnimationTrack](#)

[OpenScadCodePart](#)

[OpenScadCodePartKind](#)

[OpenScadDocument](#)

[OpenScadGenerationResult](#)

[OpenScadNode](#)

[OpenScadNodeCategory](#)

[OpenScadNodeDefinition](#)

[OpenScadParameterDefinition](#)

[OpenScadParameterType](#)

[OpenScadValidationIssue](#)

[OpenScadValidationResult](#)

[OpenScadValue](#)

[OrganicConnectionRuntimeSnapshot](#)

[OrganicConnectionState](#)

[OrganicPluginOptions](#)

[OrganicPluginWorkItem](#)

OrganicProtocolProfile	PublicationImportResult
OrganicProtocolSettings	PublicationInteraction
OrganicReplayPolicySnapshot	PublicationInteractionAction
OrganicTextInsertionProposal	PublicationLanStreamingSettings
OrganicWorkDecisionRequest	PublicationLayerItem
PagePreset	PublicationLayerMove
PaintPictureLayer	PublicationLayerOrderRequest
PanelComponentToolDescriptor	PublicationLayoutConstraintRequest
PanelElement	PublicationLiveSourceFitMode
PanelTextPatternStoreOptions	PublicationLiveSourceKind
PictureBlendMode	PublicationMapPoint
PictureDocument	PublicationMediaElement
PictureDrawTool	PublicationMediaPlaybackTrigger
PictureEditorResult	PublicationMediaSegment
PictureFillKind	PublicationMenuDestinationKind
PictureImportResult	PublicationMenuItem
PictureLayer	PublicationMenuSourceMode
PictureLayerKind	PublicationPage
PicturePoint	PublicationPageTransition
PictureRasterFitMode	PublicationPageTransitionKind
PictureRenderKind	PublicationPanelLayoutMode
PictureShapeKind	PublicationPanelNavigationMode
PictureStroke	PublicationPanelPresetDescriptor
PictureStrokeKind	PublicationPanelView
PictureTextAlignment	PublicationPlaybackSettings
PieChartStyle	PublicationPoint
PlatformChatMessage	PublicationProjectSettings
PolarChartStyle	PublicationRecordingSettings
PolygonResampleRequest	PublicationShape
PublicationAnimation	PublicationStreamAudioCodec
PublicationAnimationDirection	PublicationStreamingHotkey
PublicationAnimationEasing	PublicationStreamingSettings
PublicationAnimationEffect	PublicationStreamOutput
PublicationAnimationPhase	PublicationStreamProvider
PublicationAnimationTrigger	PublicationStreamQualityPreset
PublicationAudioDisplayKind	PublicationStreamRecordingVariant
PublicationBarcodeErrorCorrection	PublicationStreamSessionMode
PublicationBarcodeFormat	PublicationStreamTransport
PublicationBarcodeModuleShape	PublicationStreamVideoCodec
PublicationCanvasBounds	PublicationVectorFeatureKind
PublicationCanvasZoomMode	PublicationVectorMapBaseLayer
PublicationCaptureBackend	PublicationVectorMapFeature
PublicationChatDisplayMode	PublicationVideoFitMode
PublicationChatPlatform	PublicationViewSettings
PublicationComponentAction	PublicationWebBinding
PublicationComponentActionKind	PublicationWebHeader
PublicationComponentActionTrigger	PublicationWebHttpMethod
PublicationComponentConnection	PublicationWebResponseFormat
PublicationComponentDataMode	PublicationWebTransportKind
PublicationComponentEditMode	PublisherDocumentationComment
PublicationComponentEditorKind	PublisherDocumentationManifest
PublicationComponentField	PublisherDocumentationProfile
PublicationComponentFieldArea	PublisherDocumentationStatus
PublicationComponentHttpMethod	PublisherDocumentationViewerRequest
PublicationComponentKind	PublisherDocumentationViewerState
PublicationComponentPanel	PublisherDocumentDefaultsPolicy
PublicationComponentProcessingMode	PublisherMediaSessionDefaultsPolicy
PublicationComponentScope	PublisherNativeInteropPolicy
PublicationComponentSelectionMode	PublisherPictureStudioPolicy
PublicationComponentSummaryType	PublisherRegexPolicy
PublicationConnectorPort	PublisherRuntimeCollection
PublicationContentFitMode	PublisherRuntimePattern
PublicationDataColumn	PublisherRuntimePolicyOptions
PublicationDataObject	PublisherRuntimePolicySnapshot
PublicationDataRow	PublisherRuntimeValueStoreOptions
PublicationDataSourceKind	PublisherStudioConfigurationDocument
PublicationDataValueKind	PublisherStudioConfigurationNode
PublicationDocument	PublisherStudioPathOptions
PublicationElement	PublisherTwitchConfiguration
PublicationElementKind	PublisherTwitchEndpointPolicy
PublicationElementTemplate	RasterPictureLayer
PublicationFieldRecord	RecurringScreenReaderSession
PublicationGridRow	RenderExportCapability
PublicationHtmlExportSupport	RenderPictureLayer

ScreenshotCompletion	TextFrameElement
ServiceApiDescriptor	TextPictureLayer
ShapeElement	TwitchDeviceAuthorization
ShapePictureLayer	TwitchIngestCandidate
SignalCompletionAction	TwitchOAuthConnectionResult
SignalConnectorSettings	UserNotificationMessage
SignalConnectorTrigger	UserNotificationSeverity
SignalConnectorVisual	VideoEffectBlendMode
SignalGesture	VideoEffectFilter
SparklineChartStyle	VideoEffectFilterKind
SpreadsheetElement	VideoEffectLayer
SpreadsheetStorageFormat	VideoEffectLayerKind
StoryPageLayout	VideoElement
StoryStorageFormat	VideoFrameRegion
StreamingChatSendResult	VideoLayerMainframeInsertRequest
StreamingDeviceProfile	VideoProjectDocument
StreamingHardwareEncoderPreference	VideoProjectImportResult
StreamingMachineSettings	WordArtElement
StreamingProviderAuthenticationMode	WordArtFillKind
StreamingProviderProfile	WordArtPathPoint
StreamingSessionSnapshot	WordArtWarp
SvgPictureLayer	

PublisherStudio.BusinessObjects.Diagnostics

Namespace (1)
PublisherStudio.BusinessObjects.Diagnostics
Type (1)
DebugExceptionDiagnosticsOptions

PublisherStudio.BusinessObjects.Streaming

Namespace (1)	
PublisherStudio.BusinessObjects.Streaming	
Type (7)	
ChatSendRequest	MediaHotkey
IngestAnnouncement	NativeCaptureRequest
MediaHostHotkeyEvent	ProgramPageRequest
	ToggleRequest

PublisherStudio.Components.Editor

Namespace (1)	
PublisherStudio.Components.Editor	
Type (5)	
PictureEditor	SpreadsheetDataSelection
SpreadsheetDataObjectResult	SpreadsheetEditorResult
	SvgWordArtText

PublisherStudio.Components.Shared

Namespace (1)
PublisherStudio.Components.Shared
Type (1)
OperationalErrorBoundary

PublisherStudio.Controllers

Namespace (1)	
PublisherStudio.Controllers	
Type (20)	
ApiSurfaceCatalogService	AssetController

AutomationInputController
AutomationRuntimeController
AutomationScreenshotController
CodeController
ConfigurationController
DocumentationController
DomainContextController
MediaConversionController
OpenScadController

OrganicPluginController
OrganicWireHttpController
PublicationController
PublicationLayoutController
RenderExportController
RuntimePolicyController
SpreadsheetController
VideoLayerInterchangeController
WebDataController

PublisherStudio.Controllers.Streaming.UseCases

Namespace (1)

PublisherStudio.Controllers.Streaming.UseCases

Type (6)

NativeCaptureController
StreamingChatController
StreamingIngestController

StreamingLanController
StreamingRuntimeController
StreamingSessionController

PublisherStudio.Diagnostics

Namespace (1)

PublisherStudio.Diagnostics

Type (3)

ControllerRequestLoggingFilter

DebugFirstChanceExceptionLoggingHostedService
ServiceRegistrationLoggingHostedService

PublisherStudio.HostedServices.OrganicPlugins

Namespace (1)

PublisherStudio.HostedServices.OrganicPlugins

Type (1)

LocalGptDiscoveryHostedService

PublisherStudio.HostedServices.Streaming

Namespace (1)

PublisherStudio.HostedServices.Streaming

Type (2)

GlobalHotkeyHostedService

TwitchOAuthMaintenanceService

PublisherStudio.Hubs.Streaming.Chat

Namespace (1)

PublisherStudio.Hubs.Streaming.Chat

Type (1)

PlatformChatHub

PublisherStudio.Hubs.Streaming.Lan

Namespace (1)

PublisherStudio.Hubs.Streaming.Lan

Type (1)

WebRtcSignalingHub

PublisherStudio.Models

Namespace (1)

PublisherStudio.Models

Type (1)

SpreadsheetEditorViewModel

PublisherStudio.Services

Namespace (1)

PublisherStudio.Services

Type (47)

ApplicationPortResolver
ConnectorGeometry
EditorStateService
IApplicationPortResolver
IPagePresetCatalog
IPublicationGridRowFactory
IPublicationMarkupService
IPublisherDocumentFactory
IRuntimeEndpointState
IRuntimeEndpointWriter
IStoryPageLayoutService
LiveDataColumn
LiveDataObjectSnapshot
LiveElementSnapshot
LivePageSnapshot
LivePublicationSnapshot
LivePublicationSummary
PagePresetCatalog
PanelStudioTextService
PictureDocumentService
PictureEditorStateService
PublicationAnimationData
PublicationComponentService
PublicationDataService
PublicationElementTraversal
PublicationFileService
PublicationGridRowFactory
PublicationLiveDataRegistry
PublicationMarkupService
PublicationMediaAssetStore
PublicationMediaData
PublicationRecoveryService
PublicationRecoverySnapshot
PublicationWebDataService
PublicationWebhookStore
PublisherDocumentFactory
PublisherStudioServiceCollectionExtensions
RichTextDocumentFactory
RuntimeEndpointState
RuntimeEndpointWriter
SpreadsheetDocumentService
SpreadsheetEditorSession
SpreadsheetSessionStore
StoryPageLayoutService
SystemFontCatalog
WebhookPayload
WordArtPathGeometry

PublisherStudio.Services.Automation

Namespace (1)

PublisherStudio.Services.Automation

Type (10)

BusinessObjectContextService
IApiSurfaceCatalogService
IBusinessObjectContextService
IScreenshotCaptureService
IServiceArchitectureRegistry
IUserInputAutomationService
ScreenshotCaptureService
ServiceArchitectureDescriptor
ServiceArchitectureRegistry
UserInputAutomationService

PublisherStudio.Services.CodeEditing

Namespace (1)

PublisherStudio.Services.CodeEditing

Type (4)

CodeFormattingService
CodeLanguageService
ICodeFormattingService
ICodeLanguageService

PublisherStudio.Services.Configuration

Namespace (1)

PublisherStudio.Services.Configuration

Type (19)

ApplicationPathService
FileLocalizationService

IApplicationPathService
IFileLocalizationService
IPanelStudioTextPatternDataService
IPublisherDxFunctionCatalogDataService
IPublisherRuntimePatternService
IPublisherRuntimePolicyDataService
IRenderExportCatalogService
ISystemVariableStoreService
OrganicConnectionRuntimeStateService

OrganicReplayPolicyDataService
PanelStudioTextPatternDataService
PublisherDxFunctionCatalogDataService
PublisherDxFunctionCatalogDocument
PublisherRuntimePatternService
PublisherRuntimePolicyDataService
RenderExportCatalogService
SystemVariableStoreService

PublisherStudio.Services.Documentation

Namespace (1)

PublisherStudio.Services.Documentation

Type (4)

IPublisherDocumentationCatalogService
IPublisherDocumentationViewerService

PublisherDocumentationCatalogService
PublisherDocumentationViewerService

PublisherStudio.Services.MediaConversion

Namespace (1)

PublisherStudio.Services.MediaConversion

Type (2)

IMediaConversionService

MediaConversionService

PublisherStudio.Services.MediaStudio.UseCases

Namespace (1)

PublisherStudio.Services.MediaStudio.UseCases

Type (1)

MediaTimelineEditService

PublisherStudio.Services.OpenScad

Namespace (1)

PublisherStudio.Services.OpenScad

Type (16)

IOpenScadCatalogService
IOpenScadDocumentService
IOpenScadNodeFactoryService
IOpenScadNodeRenderer
IOpenScadValueFormatter
IOpenScadVideoLayerAdapter
OpenScadCatalogService
OpenScadDocumentService

OpenScadModuleCallNodeRenderer
OpenScadNodeFactoryService
OpenScadPrimitiveNodeRenderer
OpenScadRawNodeRenderer
OpenScadRenderContext
OpenScadValueFormatter
OpenScadVideoLayerAdapter
OpenScadWrapperNodeRenderer

PublisherStudio.Services.OrganicPlugins

Namespace (1)

PublisherStudio.Services.OrganicPlugins

Type (29)

ILocalGptConnectionService
ILocalGptDiscoveryRegistry
IOrganicCapabilityCatalog
IOrganicConnectionRuntimeState
IOrganicPermissionStore
IOrganicPluginProtocolCodec
IOrganicReplayGuard
IOrganicReplayPolicyDataService
IOrganicResultStore
IOrganicRuntimeSecurityService

IOrganicTransportSecurityPolicy
IOrganicWireEnvelopeFactory
IOrganicWorkCoordinator
IOrganicWorkExecutor
IRecurringScreenReaderService
LocalGptConnectionService
LocalGptDiscoveryRegistry
ObjectStoreOrganicCapabilityCatalog
OrganicCapabilityCatalog
OrganicPermissionStore

OrganicPluginProtocolCodec
OrganicReplayGuard
OrganicResultStore
OrganicRuntimeSecurityService
OrganicTransportSecurityPolicy

OrganicWireEnvelopeFactory
OrganicWorkCoordinator
OrganicWorkExecutor
RecurringScreenReaderService

PublisherStudio.Services.Panels

Namespace (1)

PublisherStudio.Services.Panels

Type (1)

PanelDocumentService

PublisherStudio.Services.PictureStudio.Import

Namespace (1)

PublisherStudio.Services.PictureStudio.Import

Type (2)

OpenRasterImportService

SvgInterchangeSanitizer

PublisherStudio.Services.Publication

Namespace (1)

PublisherStudio.Services.Publication

Type (2)

IPublicationElementLayoutService

PublicationElementLayoutService

PublisherStudio.Services.Publication.Import

Namespace (1)

PublisherStudio.Services.Publication.Import

Type (1)

OpenDocumentImportService

PublisherStudio.Services.Streaming.Capture

Namespace (1)

PublisherStudio.Services.Streaming.Capture

Type (11)

DiscoveredNativeMediaDeviceInfo
INativeCaptureSessionFactory
IWindowsProcessLoopbackCapture
IWindowsProcessLoopbackCaptureFactory
IWindowsProcessLoopbackNativeService

NativeCaptureRegistry
NativeCaptureSession
NativeCaptureSessionFactory
NativeDeviceDiscovery
WindowsProcessLoopbackCaptureFactory
WindowsProcessLoopbackNativeService

PublisherStudio.Services.Streaming.Chat

Namespace (1)

PublisherStudio.Services.Streaming.Chat

Type (1)

PlatformChatService

PublisherStudio.Services.Streaming.Encoding

Namespace (1)

PublisherStudio.Services.Streaming.Encoding

Type (8)

EncoderOrchestrator
EncoderSessionService
FfmpegEncoderResolver
FfmpegEncoderSet

FfmpegLocator
FfmpegVideoEncoder
MediaOutputDefinition
MediaRecordingDefinition

PublisherStudio.Services.Streaming.Hotkeys

Namespace (1)

PublisherStudio.Services.Streaming.Hotkeys

Type (5)

GlobalHotkeyService
IWindowsHotkeyNativeService

WindowsHotkeyNativeMessage
WindowsHotkeyNativePoint
WindowsHotkeyNativeService

PublisherStudio.Services.Streaming.Lan

Namespace (1)

PublisherStudio.Services.Streaming.Lan

Type (4)

LanStreamingServer
MediaLanDefinition

RtspLanServer
WebRtcSignalingService

PublisherStudio.Services.Streaming.MediaHost

Namespace (1)

PublisherStudio.Services.Streaming.MediaHost

Type (4)

MediaHostOutputRequest
MediaHostSessionResponse

MediaHostStartSessionRequest
StreamingMediaHostClient

PublisherStudio.Services.Streaming.Metadata

Namespace (1)

PublisherStudio.Services.Streaming.Metadata

Type (1)

NowPlayingReader

PublisherStudio.Services.Streaming.OAuth

Namespace (1)

PublisherStudio.Services.Streaming.OAuth

Type (1)

TwitchOAuthService

PublisherStudio.Services.Streaming.Profiles

Namespace (1)

PublisherStudio.Services.Streaming.Profiles

Type (1)

StreamingProfileStore

PublisherStudio.Services.Streaming.Sessions

Namespace (1)

PublisherStudio.Services.Streaming.Sessions

Type (8)

ILanStreamingServerFactory
IMediaSessionFactory
IPlatformChatServiceFactory
LanStreamingServerFactory

MediaSession
MediaSessionFactory
MediaSessionRegistry
PlatformChatServiceFactory

PublisherStudio.Services.Streaming.Settings

Namespace (1)

PublisherStudio.Services.Streaming.Settings

Type (1)

PublicationStreamingSettingsStore

PublisherStudio.Services.Streaming.UseCases.Chat

Namespace (1)

PublisherStudio.Services.Streaming.UseCases.Chat

Type (3)

IStreamingChatResultFactory
StreamingChatResultFactory
StreamingChatUseCases

PublisherStudio.Services.Streaming.UseCases.Editor

Namespace (1)

PublisherStudio.Services.Streaming.UseCases.Editor

Type (1)

StreamingSessionService

PublisherStudio.Services.Streaming.UseCases.Ingest

Namespace (1)

PublisherStudio.Services.Streaming.UseCases.Ingest

Type (1)

StreamingIngestUseCases

PublisherStudio.Services.Streaming.UseCases.Lan

Namespace (1)

PublisherStudio.Services.Streaming.UseCases.Lan

Type (3)

StreamingAsset
StreamingLanStatus
StreamingLanUseCases

PublisherStudio.Services.Streaming.UseCases.NativeCapture

Namespace (1)

PublisherStudio.Services.Streaming.UseCases.NativeCapture

Type (1)

NativeCaptureUseCases

PublisherStudio.Services.Streaming.UseCases.Runtime

Namespace (1)

PublisherStudio.Services.Streaming.UseCases.Runtime

Type (2)

StreamingRuntimeCapabilities

StreamingRuntimeUseCases

PublisherStudio.Services.Streaming.UseCases.Sessions

Namespace (1)

PublisherStudio.Services.Streaming.UseCases.Sessions

Type (1)

StreamingSessionUseCases

PublisherStudio.Services.UserExperience

Namespace (1)

PublisherStudio.Services.UserExperience

Type (2)

IUserNotificationService

UserNotificationService

PublisherStudio.Services.VideoStudio.Export

Namespace (1)

PublisherStudio.Services.VideoStudio.Export

Type (6)

BrowserRuntimeTemplateService
IBrowserRuntimeTemplateService
IPolygonGeometryService

IVideoLayerInterchangeService
PolygonGeometryService
VideoLayerInterchangeService

PublisherStudio.Services.VideoStudio.Import

Namespace (1)

PublisherStudio.Services.VideoStudio.Import

Type (1)

VideoProjectImportService

The owner release build regenerates this handbook from the same DocFX HTML payload before packaging each runtime.